

Reference A of CHAPTER 1

Asset Classes and Financial Instruments

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Part 1

Investment and Crisis

Investments

- What is investments?
 - The **current** commitments of money or other resources in the expectation of reaping **future** benefits.
 - To sacrifice something of value now expecting to benefit from that sacrifice later.
- Essential nature of investment
 - Reduced current consumption
 - Planned later consumption

Real Assets VS Financial Assets (1)

- Real Assets
 - Assets used to produce goods and services (productive capacities)
 - E.g., land, buildings, machines, knowledge
- Financial Assets
 - Do not contribute directly to the productive capacity of the economy
 - Claims (rights) to the income generated by real assets (or government)

CONCEPT CHECK

Are the following assets real or financial?

1. Patents
2. A college education
3. A \$10 bill

Real Assets VS Financial Assets (2)

- Example

If we cannot own our own auto plant(real asset), we can buy shares in HXX Motor (financial asset).

→ We get the income derived from the production of automobiles from this investment.

- The success or failure of financial assets depends on the **underlying** real assets.

Financial Assets (1)

- Fixed-income (or debt) securities
 - Promise either a fixed stream of income or a stream of income that is determined according to a specified formula
 - Bonds: corporate bonds, Treasury bills, Treasury bonds etc.
 - Money market VS capital market: short term VS long term (e.g. Treasury bill or CD (certificates of deposit) VS Treasury bond)

Q. What is the difference between the followings?
Treasury bill, Treasury note, Treasury bond

Financial Assets (2)

- Common stock (Equity)
 - Ownership share in the corporation
 - Equity holders are not promised any particular payment.
 - Tied to the success of the firm and its real assets.
 - Equity investments tend to be riskier than investments in fixed-income securities.
- Common stock VS preferred stock

Financial Assets (3)

- Derivative securities
 - Contracts provide payoffs that are determined by the prices of other assets (**underlying** assets) such as bond and stock prices.
 - Example: call option on a share of Samsung Elec. stock.
 - Have become an essential part of the investment environment.
 - Hedge risks or transfer them to other parties.

Financial Markets and the Economy: The Roles of Financial Assets (1)

- The Information Role of Financial Markets
 - Play a central role in the allocation of capital resources. (E.g., good vs bad prospect)
 - Example 1: A stock price reflects the future success or failure of the firm.
 - Example 2: Wasteful capital allocation (e.g., short term rushes for ventures)
- Consumption Timing
 - People can store (or invest) their wealth in financial assets (e.g., bank accounts) for their future consumptions.
 - Example: Retiree spend more than they currently earn. How they can shift their purchasing power?

Financial Markets and the Economy: The Roles of Financial Assets (2)

- Allocation of Risk
 - Virtually all real assets involve some risks.
 - Example: Samsung builds a new semiconductor plant by selling bonds and stocks to the **public**. (conservative VS risk-tolerant investors)
 - Beneficial to both investors and firms
 - Firms can raise capital to finance their investments.
- Separation of Ownership and Management
 - With global markets and large-scale production, the size and capital requirements of firms have skyrocketed.
 - Owners (stock holders) VS managers : if stock holders decide sell some shares to other investors, there is no impact on the management of the firm.

Financial Markets and the Economy: The Roles of Financial Assets (3)

– Agency problem

Q. Do managers really attempt to maximize firm value? (**Moral hazard**)

- Compensation plan (e.g., executive stock option)
- The boards of directors can force out managers. (E.g., Disney, Boeing, Fennie Mae)
- Outsiders such as security analysts, and large institutional investors (e.g., pension funds) make poor performers (e.g., fund managers) uncomfortable.
- Bad performers are subject to the threat of takeover. (e.g., HP proxy fight)

Corporate Governance and Corporate Ethics

- There must be an acceptable level of transparency that allows investors to make well-informed decisions.
- Examples
 - Accounting Scandals
 - Enron and WorldCom's bankruptcy
 - Analyst Scandals
 - Citigroup's Salomon Smith Barney
 - Initial Public Offerings
 - Credit Swiss First Boston
- The **Sarbanes-Oxley Act** in the U.S.: tighten the rules of corporate governance(e.g., more independent directors).

Markets are competitive (1)

- ‘There is no free lunches.’
- The **risk-return trade-off**
 - If you want higher expected returns, you will have to pay a price in terms of accepting higher investment risk.
 - Important questions
 - Q1.** How should one measure the risk of an asset?
 - Q2.** What should be the quantitative trade-off between risk and expected return?
 - Diversification: To hold many assets.
 - Modern portfolio theory (H.Markowitz and W.Sharpe)

Markets are competitive (2)

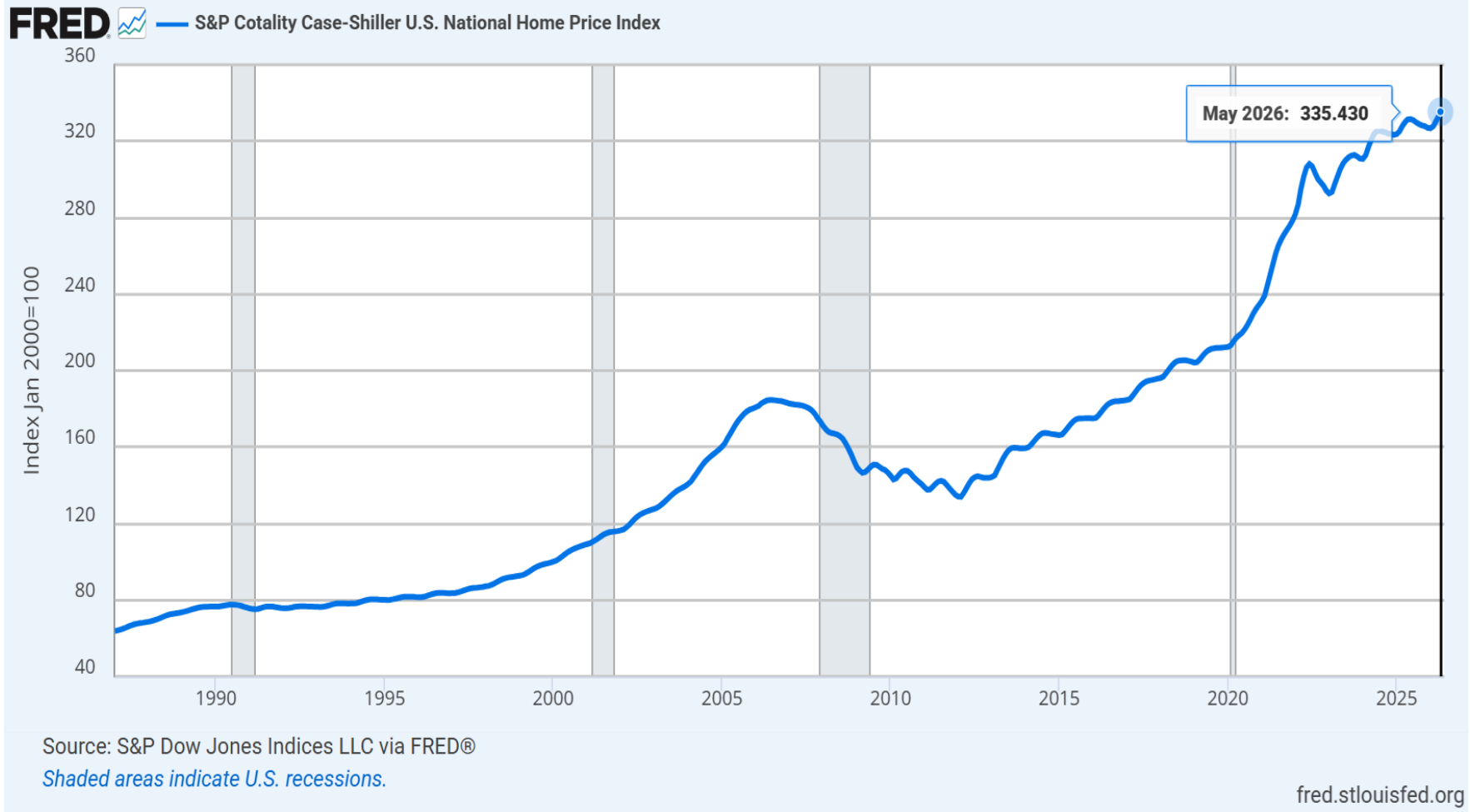
- Efficient Markets
 - **Efficient market hypothesis** (EMH): Financial markets process all relevant information about securities quickly and efficiently.
- Active Management
 - Finding undervalued securities
 - Timing the market (e.g., increase of stock investment in a bullish market)
- Passive Management
 - No attempt to find undervalued securities
 - No attempt to time
 - Holding an efficient portfolio

The Players

- Three Major Players
 - Business firms: net borrowers
 - Households: net savers
 - Governments can be both borrowers and savers. (Tax revenues VS government expenditures)
- Financial Intermediaries
 - Examples:
commercial banks, investment companies,
insurance companies, credit unions, mutual funds,
pension funds.

- **Financial intermediaries** issue their own securities to raise funds to purchase the securities of other corporations.
- **Commercial banks** raise funds by borrowing (taking deposits) and lending that money to other borrowers.
- **Investment companies** pool and manage the money of many investors, and use economies of scale and scope.
- **Investment bankers** (Goldman Sachs, Merrill Lynch, Citigroup etc) advise the issuing corporation on the prices it can charge for the securities issued, appropriate interest rates, and so forth. Handle the marketing of the security in the primary market.

The Financial Crisis of 2008 (1)



The Financial Crisis of 2008 (2): Changes in Housing Finance

- Prior to 1970, most mortgage loans would come from lender such as a neighborhood savings banks or credit unions.
- Changes begin when Fannie Mae (FNMA, Federal National Mortgage Association) and Freddie Mac (FHLMC, Federal Home Loan Mortgage Corporation) begin buying mortgage loans from originator and bundling them into large pools that could be traded like any other financial assets. (called **securitization**)
 - low-risk conforming mortgages
- Securitization by private firm of non-conforming subprime loans with higher default risk → subprime crisis!

The Financial Crisis of 2008 (3)

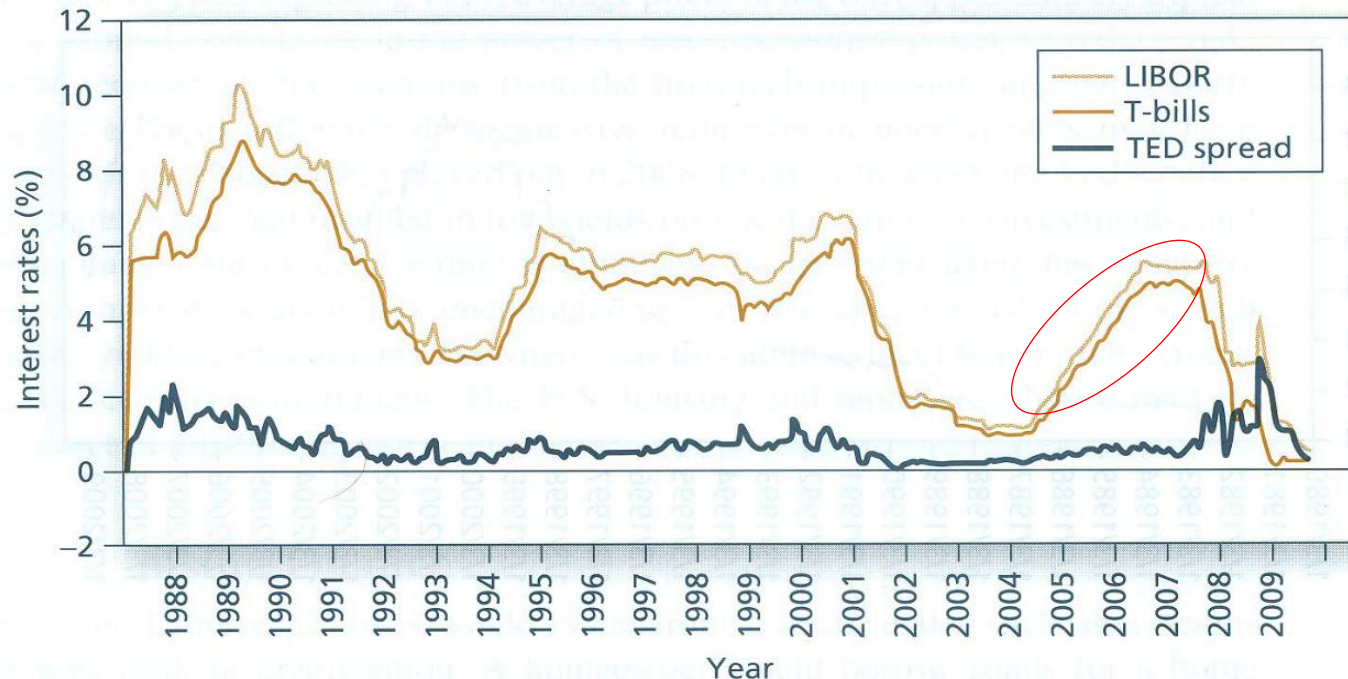
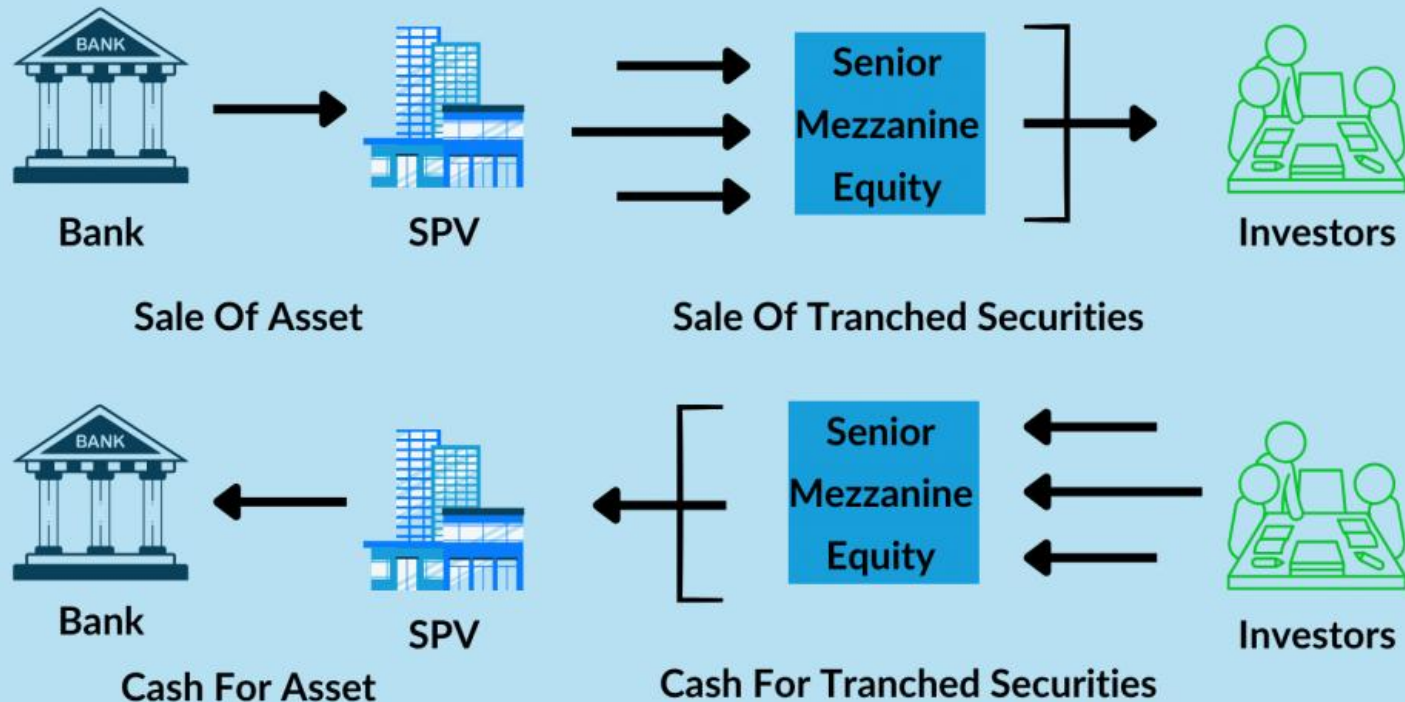


Figure 1.1 Short-term LIBOR and Treasury-bill rates and the TED spread

* TED spread = Treasury rate – LIBOR

The Financial Crisis of 2008 (4): Mortgage Derivatives

Structure Of Collateralized Debt Obligations (CDOs)



The Financial Crisis of 2008 (5): The Rise of Systemic Risk

- **Systemic risk**: a potential breakdown of the financial system when problems in one market spill over and disrupt others
- Example: AIG's insolvency
 - triggered the insolvency of other firms, particularly banks, which had relied on its promise of protection against defaults of hundreds of billions of dollars of mortgage loans.
- Recovery from the Crisis
 - The government tried to reduce risk of the financial sector by pouring capital into precarious banks. (QE/Quantitative Ease)
 - The government tries to make a centralized exchanges, in which trade some standardized OTC financial products such as CDS contracts. (CCP/central counterparty)
 - Risk-based incentive system

Part 2

Asset Classes and Products

The Money Market (1)

- Money market: short-term, marketable, liquid, and low-risk debt securities
 - Money market instruments sometimes are called *cash equivalents* or *cash* for short
 - Examples: certificate of deposit(CD), commercial paper(CP)
- Capital market: long-term, riskier securities
 - Longer-term bond market, equity markets, derivative markets for options and futures

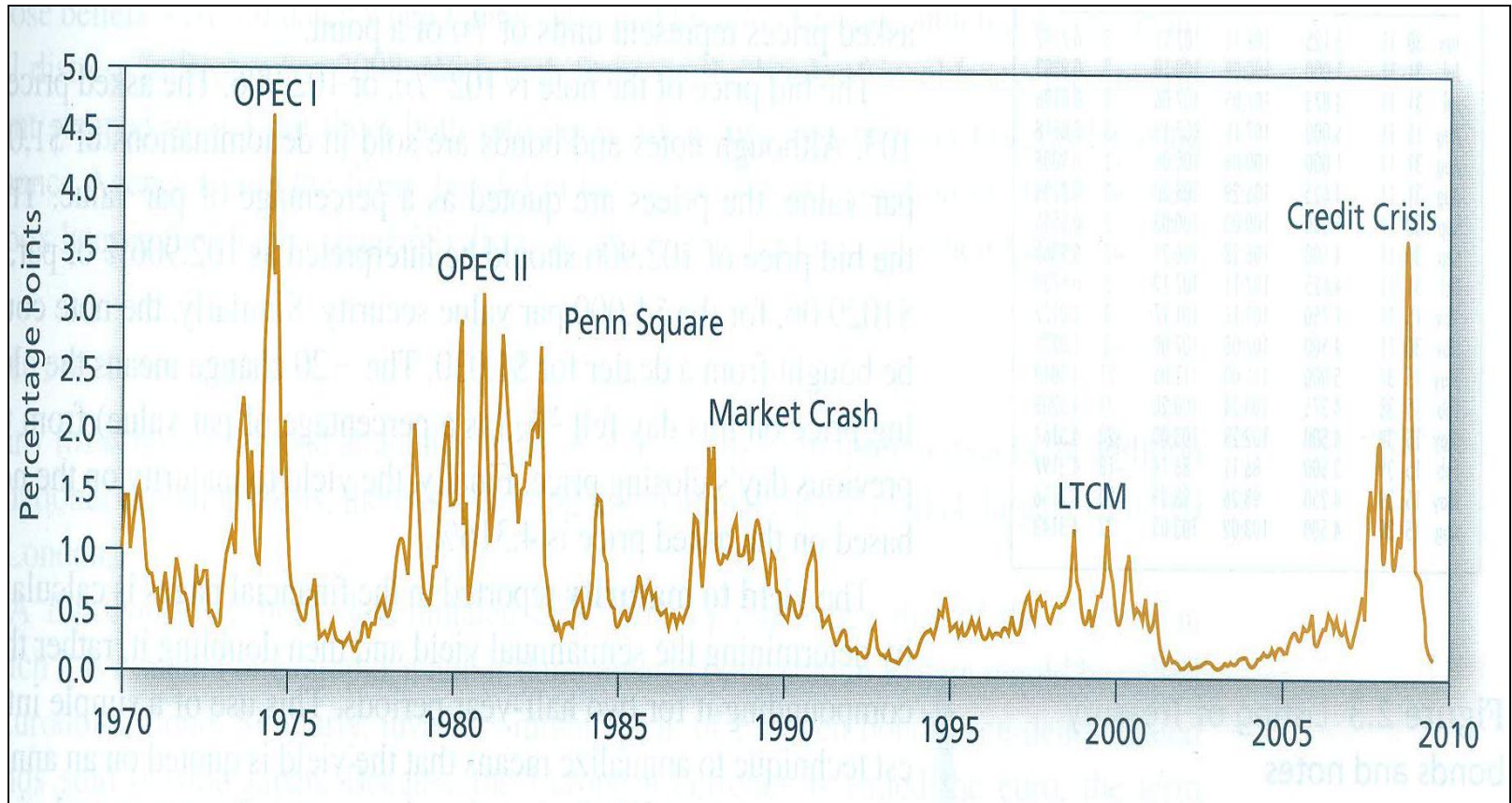
The Money Market (2)

- Treasury bills (**T-bills**, bills)
 - Issued by the U.S. government with initial maturities of 4, 13, 26, or 52 weeks.
 - Individuals can buy T-bills on the secondary market from a government security dealer.
 - T-bill sells in minimum denominations of only \$100. (\$10,000 common)
 - T-bills is exempt from all state and local taxes.
- CD(Certificate of Deposit)
 - *A time deposit (term deposit, fixed deposit)* with a bank (may not be withdrawn on demand)
 - The bank pays interest and principal to the depositor only at the end of the fixed term of the CD.

The Money Market (3)

- CP(Commercial Paper)
 - A short-term unsecured debt notes usually issued by large, well-known companies.
 - A sharp increase in asset-backed commercial paper (ABCP) issued by financial firms such as banks have been observed in recent years. It was short-term CP typically used to raise funds for the institution to invest in other assets (collateral).
- Repurchase agreements (Repo, **RP**)
 - The dealer sells government securities to an investor on an overnight basis, with an agreement to buy back those securities some future day (in general, the next day) at a slightly higher price.
- **KOFR, SOFR**

Figure. The spread between 3-month CD and Treasury bill rates



The Bond Market (1)

- Treasury Notes and Treasury Bonds
 - Maturities: ~10 years(**T-note**), 10~30 years (**T-bond**)
 - Make semiannual interest payments(coupon payments)
- Inflation-Protected Treasury Bonds
 - Treasury Inflation-Protected Securities (**TIPS**): The principal amount on these bonds is adjusted in proportion to increases in the Consumer Price Index (CPI).
- Corporate Bonds
 - Issued by private firms.
 - Types: secured bond (collateral-backed), debenture (no collateral), subordinated debenture (lower-priority)
 - Semiannual interest payments.
 - Subject to larger default risk than government securities.
 - Options in corporate bonds
 - Callable bond: give the firm the option to repurchase the bond from the holder at a stipulated call price.
 - Convertible bond(**CB**): give the bondholder the option to convert each bond into a stipulated number of shares of stock.

Equity Securities

- Common stock (Equity) as ownership shares
 - Each share entitles its owner to one vote on any matters of corporate governance and to a share in financial benefits of ownership.
 - Proxy fight (proxy contest)
 - Agency problem: shareholders VS managers
 - *Residual claim* : Stock holders are the last in line of all those who have a claim on the assets and income of the corporation.
 - *Limited liability* : The most stock holders can lose in the event of failure of the corporation is only their original investment.
- ✓ Preferred stock is a hybrid type of equity ownership in a corporation that combines features of both common stock and bonds, granting shareholders a higher priority claim on dividends and assets.

Stock Market Indexes (1)

- Factors for Construction of Stock Indexes
 - Representative?
 - Broad or narrow?
 - How is it weighted?
- Dow Jones Industrial Averages (DJIA)
 - 30 blue-chip corporations (since 1896)
 - Price-weighted average → Overweights the performance of high-priced stock
- Standard & Poor's Composite 500 (S&P 500)
 - Improvement over DJIA: broadly based index of 500 firms
 - market-value-weighted → unaffected by stock splits.
 - Individuals can buy shares in mutual fund or exchange-traded fund (ETF)

Stock Market Indexes (2)

- Kospi / Kospi 200
- Nikkei 225 (Japan)
- FTSE (Financial Times of London)
- Dax (Germany,
- Hang Seng (Hong Kong)
- TSX (Canada)

Derivatives Markets

- **Derivatives** = Derivative assets
= Contingent claims: Their values are derived from or contingent on the values of other assets (underlying assets).
- Options: call VS put
 - Expiration date (expiry, maturity)
 - Exercise price (strike price)
 - Option prices increase with time until expiry.
- Futures contracts (futures)
 - Calls for delivery of an asset at delivery date (maturity) for an agreed-upon price (future price).
 - Long position (buy) VS short position (sell)
- Options VS Futures
 - Option premium VS no cost
 - Composition of options (**How?**) → futures
- Others: swaps, swaptions etc.